

Track 3 — Java + AI/RAG Differentiator

This moves you from regular Java Lead to Java + AI/RAG + architecture candidate. It is for stronger roles, better rates and long-term market value — not last-minute panic.

• Goal & Positioning

POSITIONING STATEMENT

“My core is Java / Spring Boot microservices. AI/RAG is an extension layer I am building on top of backend systems — ingest, chunk, embed, retrieve, augment and generate grounded answers. I frame it honestly as a growing differentiator, not as inflated AI production experience.”

- ☐ Build small **finished** proof projects, not big unfinished demos.
- ☐ Use AI/RAG to strengthen eCommerce and backend positioning.
- ☐ Publish GitHub / LinkedIn / Medium proof gradually.
- ☐ Do not oversell production AI you have not built in production.

THE BAR

One small, **finished**, public artifact with a clean README beats four half-written drafts. Finished and small > ambitious and abandoned. Ship one thing, then the next.

• The 6-Month Roadmap

Month 1 — AI/RAG Foundations

Build vocabulary and an honest understanding of GenAI for backend roles.

STUDY

- ☐ Tokens, context window, temperature
- ☐ Embeddings, vector search
- ☐ Semantic search, chunking
- ☐ Retrieval quality

OUTPUT TO FINISH

- ☐ Explain the RAG pipeline in 2 min: ingest → chunk → embed → retrieve → augment → generate
- ☐ Explain hallucination and grounding

Month 2 — Spring AI Practical Layer

Build one small, finished RAG project on Spring Boot.

STUDY

- ☐ ChatClient, prompt templates
- ☐ Embedding model, pgvector
- ☐ Vector-store abstraction
- ☐ Evaluation, guardrails, security

OUTPUT TO FINISH

- ☐ **Working mini app:** upload → chunk → embed → ask → **cited answer**
- ☐ A small evaluation set + a clean README

Month 3 — eCommerce AI Use Cases

Connect AI/RAG to your eCommerce profile.

STUDY

- ☐ Gift reminders, occasion intelligence
- ☐ Product classification
- ☐ Recommendations, search relevance

OUTPUT TO FINISH

- ☐ One small gift-reminder or product-classification demo
- ☐ Frame honestly: core is Java/Spring, AI is the edge

Month 4 — Advanced Event-Driven

Go deeper in Kafka and distributed processing.

STUDY

- ☐ Ordering, retries, DLQ
- ☐ Schema evolution, event versioning
- ☐ Outbox, idempotency, observability

OUTPUT TO FINISH

- ☐ **Finish a Kafka retry/DLQ demo**
- ☐ One post on event-driven trade-offs

Month 5 — Data Modeling / Distributed Systems

Strengthen architecture depth.

STUDY

- ☐ Cassandra, consistency models
- ☐ Caching, sharding
- ☐ Backpressure, distributed locks

OUTPUT TO FINISH

- ☐ Cassandra data-model examples
- ☐ A scalable checkout/order read path

Month 6 — Public Proof

Create evidence recruiters and clients can see.

STUDY

- ☐ GitHub, README, diagrams
- ☐ Posts, mock stories
- ☐ Resume alignment

OUTPUT TO FINISH

- ☐ Publish 2–3 finished artifacts
- ☐ Record 2-minute explanations

• AI/RAG Interview Question Bank

Drill as spoken answers, framed at your honest level.

- | | | |
|-------------------------------------|--|--|
| 1 What is RAG? | 12 What is temperature? | 21 How would you build document Q&A? |
| 2 What are embeddings? | 13 What is a context window? | 22 How would you build product recommendation? |
| 3 What is vector search? | 14 How do you secure GenAI apps? | 23 How would you build a gift-reminder AI? |
| 4 Semantic vs keyword search. | 15 How do you avoid sending PII? | 24 How do you monitor LLM usage? |
| 5 What is chunking? | 16 How would you add AI to a Spring Boot app? | 25 How do you cache LLM responses? |
| 6 How do you decide chunk size? | 17 What is Spring AI ChatClient? | 26 How do you handle prompt injection? |
| 7 What is top-k retrieval? | 18 How does a vector-store abstraction work? | 27 How do you version prompts? |
| 8 What is reranking? | 19 pgvector vs a dedicated vector DB (high-level). | 28 How do you test AI output? |
| 9 What is hallucination? | 20 How do you evaluate RAG quality? | 29 How do you handle cost / latency? |
| 10 How do you reduce hallucination? | | 30 How do you explain AI experience honestly? |

• Portfolio Proof Checklist

- ☐ Spring AI RAG Q&A mini-project with a clean README.
- ☐ Kafka retry / DLQ / idempotent-consumer demo.
- ☐ eCommerce gift-reminder AI mini-project.
- ☐ Cassandra data-modeling notes or article.
- ☐ System-design diagram for checkout / risk evaluation.
- ☐ LinkedIn post on Kafka idempotency.
- ☐ Medium article on Cassandra-vs-RDBMS thinking.
- ☐ GitHub README polished with diagram and run steps.

• Project Ideas

- | | | |
|---|--|-----------------------------------|
| 1 Document Q&A with citations | 5 Customer-support RAG over FAQs | 9 Architecture-decision assistant |
| 2 Gift-reminder occasion classifier | 6 Fraud-explanation assistant (high-level) | 10 Resume / JD matcher |
| 3 Product classification from title/description | 7 Search-relevance re-ranker (high-level) | |
| 4 Cart-abandonment assistant | 8 Kafka event summarizer | |

• What Not To Claim

CREDIBILITY GUARDRAIL

This protects you from getting dismantled on the one area you can't fully back. Keep it strict.

- ☐ Do not claim **production GenAI ownership** if it was only a demo.
- ☐ Do not claim **model training** unless you actually trained / fine-tuned a model.
- ☐ Do not claim **vector-DB production expertise** until you build one and can explain the trade-offs.
- ☐ Say **"I built / explored / prototyped"** when that is the honest level.

• Track 3 Coverage Map

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|---|---|
| ☐ Explain RAG clearly in 2 minutes. | ☐ Explain Kafka retries, DLQ, outbox and idempotency deeply. |
| ☐ Explain embeddings and vector search. | ☐ Explain Cassandra data-modeling basics. |
| ☐ Describe Spring AI components at a high level. | ☐ Have at least 2 finished GitHub demos or diagrams. |
| ☐ Built or can whiteboard a document-Q&A RAG system. | ☐ Have a few public posts or solid drafts. |
| ☐ Built or can whiteboard an eCommerce AI use case. | ☐ Position AI honestly, without overselling. |